

Psychological state analysis of social media users based on language model and text representation similarity

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Abstract

Although studies on the classification and explanation of user mental states have been conducted by using social media data, while the method based on the latest language model achieves considerable classification performance on some datasets, individuals belonging to the same diagnostic category are still heterogeneous from each other under the existing clinical diagnostic criteria. Failure to identify the source of the difference may result in "non-targeted" and ineffective intervention. Refer to Research Domain Criteria (RDoC), the new research framework from the National Institute of Mental Health (NIMH), this work models the mental state of users as an intervention dimension based on their texts. Specifically, targeting data on self-reported depressed users of social media, this work analyses potential motivational and emotional states expressed by user texts in two ways based on the language model and looks into the differences of distributions in motivational and emotional states between depressed and nondepressed users. The results show that universal language models have been able to identify differences in language patterns among people with different mental health conditions to a certain extent, consider how this can be used to further explore psychological process modeling in longitudinal user data.

1 Introduction

Mental distress can be defined as internal signals of anxiety or stress that occur when a person faces challenges. Reactions to ordinary life events can lead to significant difficulties with emotions, cognition, physical symptoms, and behavior. These mental health challenges often do not require treatment but will respond to support or guidance from significant others.

Mental disorders are health conditions characterized by significant dysfunction in emotions, cognition, physical states, and/or behaviors. When these dysfunctions co-occur in specific constellations, persist for an extended period, or exceed some threshold level of intensity and frequency, they can impair a person's overall functioning and may require treatment by trained professionals (Beyer and Boazak, 2021).

Often, symptoms of mental health disorders develop gradually over time for an individual and are only noticed when they eventually cause some significant degree of distress to the individual himself or others in his or her social network. Therefore, even regardless of socioeconomic factors, there are wide individual differences in identifying mental health problems and accessing mental health care services.

As a platform for information sharing and exchange, social media enables users to express themselves in a relatively daily state, and its content may reflect the real life and psychological state of users. The behavior of users using social media also indicates their motivation to express

and communicate, and to a certain extent, regulate their own emotions and stress in this way, especially when using social media platforms with specific themes, such as depression support forums, subreddits to discuss depression or anxiety, etc. Therefore, analyzing the content posted by users to understand their mental states, promptly detecting potential mental health problems, and providing appropriate support and interventions are of practical significance to promoting individual health and reducing the incidence of mental disorders.

Most of the current mental health studies based on social media data focus on the classification of users' mental health status, either with or without a single disorder, or to predict multiple mental disorders at the same time, and pay attention to the interpretability of the classification. There are also some studies analyzing differences in language patterns between people with mental health conditions and healthy people. Although some shared tasks at large conferences have provided mental health datasets, the data sources and collection methods of relevant studies vary, and the studies are isolated from each other and lack correlation and comparability.

Depression is a mental disorder that currently attracts much attention and is prevalent in society. Its diagnosis only requires the presence of one of two core symptoms: long-term negative mood and loss of interest in previously enjoyable activities. People with depression take longer to disengage from processing negative information and have difficulty suppressing irrelevant thoughts or shifting attention from one task to another to achieve goals. These problems can make it difficult for individuals to regulate their emotions and adapt to changing circumstances, making cognitive processes a key target in understanding and treating depression (Grahek et al., 2019).

The two core symptoms of depression are closely related to abnormalities in an individual's motivational and emotional processes. Cognitive processes interact with an individual's motivational and emotional processes to influence an individual's behavior. Through an individual's posting behavior on social media, it is assumed that the content of posts can reflect the individual's motivational and emotional processes to a certain extent. The motivational and emotional states of depressed and non-depressed people can be compared to find differences, which can provide a basis for the subsequent determination of appropriate intervention points.

This work attempts to analyze the differences in motivational and emotional distributions between users with self-reported depression and healthy ones as a preliminary way to observe users' psychological states. Intending to obtain motivation and emotion labels of users' social media posts without the involvement of mental health professionals, in consideration of the ability of current large language models to encode the semantics of texts, this work annotates a social media dataset in two ways by using language models: 1) employing an additional dataset with motivation and emotion labels, and comparing the similarity between text embeddings of the two datasets; 2) generating labels directly.

2 Related Work

In terms of social media-based detection of multiple mental disorders, there is research focusing on the correlation between linguistic markers across multiple mental health conditions, involving transfer learning between depression, stress, and suicide detection tasks (Aich and Parde, 2022). Another study proposes a two-step BERT domain adaptation model to detect signs of mental disorders in social media and use vocabulary knowledge to guide the masking process of the language model, involving the detection of anorexia, depression, and self-harm (Aragón et al.,

2023). There is also a study using symptoms from DSM-5 diagnostic criteria to simultaneously detect multiple mental disorders by identifying users' risk posts and obtaining symptom signatures in a symptom space common to multiple disorders (Chen et al., 2023).

Some works also explore the detection of conversational forms. One study has proposed treating the text classification task as a seq2seq question answering task, fine-tuning AllenAI's Unified-QA model on two large mental health datasets in a differentiated and private manner, and detecting multiple mental disorders in the form of simulated clinical consultation (Chhikara et al., 2023). There is also a study that has extended PHQ-9 questions to filter the data set to support the fine-tuning of question generation models on subreddit posts and comments, allowing the model to judge known and unknown information and generate appropriate follow-up questions to advance depression detection (Gupta et al., 2022).

In terms of interpretable predictions, some works have used information from the diagnostic criteria of DSM-5 and clinical questionnaires. One study uses a multi-task learning framework with tweet-level PHQ-9 classification as an auxiliary task to achieve tweet-level depression classification (Zirikly and Dredze, 2022). Another study uses datasets annotated with symptom correlation and state labels based on the diagnostic criteria of various mental disorders in DSM-5 and clinical questionnaires. First, a symptom correlation judgment model is trained, and then based on the symptom features extracted by the model, combine subject characteristics and symptom state characteristics to predict mental illness and explain results based on symptoms (Zhang et al., 2022). There is also a proposed Multi-Head Siamese network that can capture the semantics of text input and directly compare it with each symptom-related description from DSM-5, thereby finding symptom-related clues directly from the input to predict four mental disorders, while examining the weights learned by each symptom head, the model produces human-understandable explanations (Song et al., 2023).

Some works have explored the role of non-clinical standard psychosocial factors in predicting mental health conditions. One study uses feature-level fusion, model fusion, and task fusion strategies of emotion and personality information to predict five mental health conditions (Zanwar et al., 2023). There is also research exploring interpersonal risk factors in depression and suicide-related social media posts, creating a Reddit dataset for classifying two types of negative emotions in posts (Garg et al., 2023). Another study constructs a dataset based on depression and suicide-related posts on Reddit to identify four related health dimensions, including physical aspect (PA), intellectual and occupational aspect (IVA), social aspect (SA), and mental and emotional aspect (SEA) (Liyanage et al., 2023). Newer work has also compared the Discourse Emotion Dynamics (UED) metric between tweets from users with multiple self-disclosed mental health conditions and tweets from control groups (Teodorescu et al., 2023).

In the analysis of individual mental states, one study has identified two kinds of emotional change moments in longitudinal user posts (Tsakalidis et al., 2022). Another study has tracked the language trajectories of risk and protective coping strategies among healthcare workers during the COVID-19 pandemic, using mixed-effects regression models to test the effects of 30 language variables, including social language, functional words, and emotional language (Ireland et al., 2022). There is also a study using methods based on similarity, textual entailment, and LLMs to identify maladaptive patterns from user texts in mental health forums (Gollapalli et al., 2023).

In terms of universal language models, there are already pre-trained models in the mental health field, MentalBERT, and MentalRoBERTa, that perform domain adaptation training based on user

posts on Reddit (Ji et al., 2022). Another study explores the ability of Sentence-BERT, Phrase-BERT, and DiscRE models to encode discourse-level embeddings and use them to predict anxiety levels (Juhng et al., 2023). There is also a study using ChatGPT to complete zero-shot classification in the task of detecting stress, depression, and suicidal tendencies based on users' social media posts (Lamichhane, 2023).

In addition, to address the issues of validity and generalization ability of social media data, there is a study using post data of users with self-reported mental disorders on Reddit at different stages to separately train classification models, and test different models' classification performances on another Twitter-based dataset (Liu et al., 2023).

3 Method

In the absence of input from professionals in the mental health field, consider using existing LLMs for annotation. Because of its training in massive amounts of text and its good processing of human language, it is assumed to have the ability to encode real semantics from various types of text. Therefore, assuming that texts with similar semantics may also have similar mental state labels, consider assigning the most similar mental state label to the user based on the similarity between the text encoding of depressed/non-depressed users and the typical text encoding of various mental state labels. In addition, prompts that directly ask an LLM to generate the motivations and emotions of the given user text are also considered.

3.1 Annotation based on text similarity

The complete operational process is shown in Figure 1.

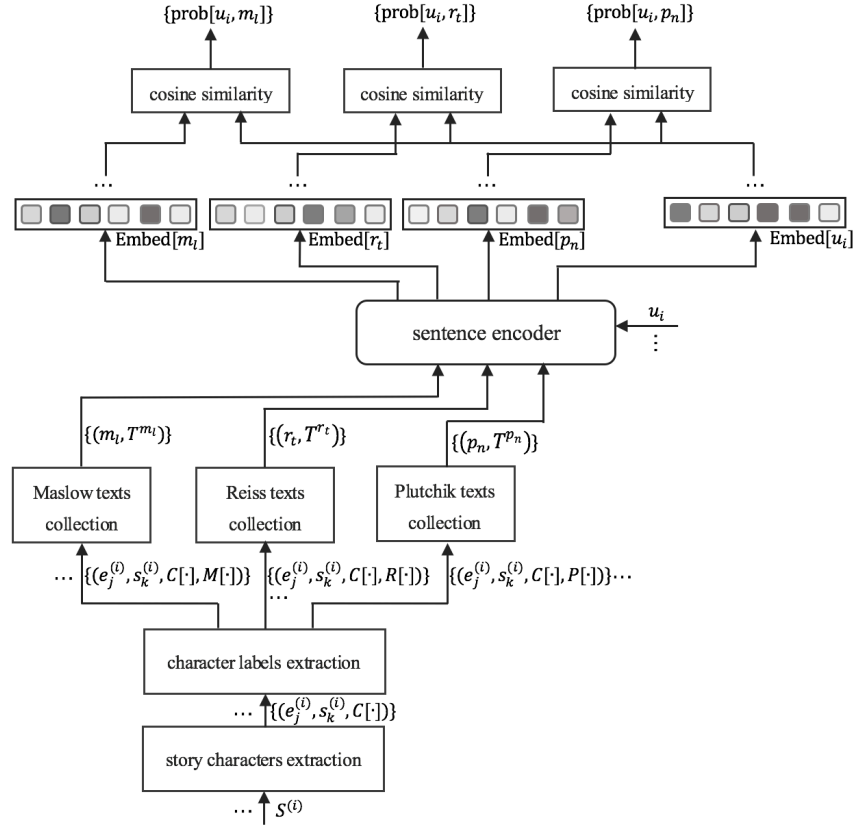


Figure 1: User annotating process based on text similarity.

3.1.1 Motivation and emotion tags

Using the storycommons dataset (Rashkiny et al., 2018), this dataset densely annotates common sense short stories based on the psychological state of the characters. The specific categories are shown in Figure 2.

1) Two popular theories of motivation, “hierarchy of needs” by Maslow (1943) and “basic motives” by Reiss (2004), are used to compile five coarse-grained (Maslow's hierarchy of needs) and nineteen fine-grained (basic) motivation categories for the convenience of annotators. Reiss tags are annotated as subcategories of Maslow categories.

2) Using the eight basic emotion dimensions from Plutchik's (1980) "wheel of emotions" because of its use as a common choice in previous literature on emotion classification.

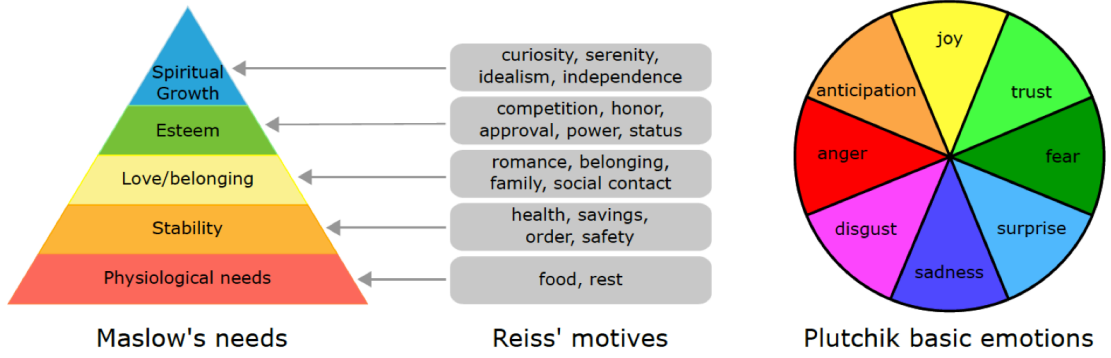


Figure 2: Theories of motivation and emotional responses (Rashkiny et al., 2018).

Maslow tag set $M = \{\text{physiological, stability, love, esteem, spiritual growth}\}$, Reiss tag set $R = \{\text{food, rest, health, savings, order, safety, romance, belonging, family, contact, competition, honor, approval, power, status, curiosity, serenity, idealism, independence}\}$, Plutchik tag set $P = \{\text{joy, trust, surprise, anticipation, fear, disgust, anger, sadness}\}$.

3.1.2 Obtaining the corresponding texts of tags

In the storycommons dataset, each story $S^{(i)}$ contains five sentences, that is $S^{(i)} =$

$\{s_1^{(i)}, \dots, s_5^{(i)}\}$, and the collection of characters in the story is represented as $E^{(i)}$. For each

character $e_j^{(i)} \in E^{(i)}$, every sentence this character appears in the story ($s_k^{(i)} \in S^{(i)}$) is annotated,

treating all sentences before this sentence in the story as context $C[e_j^{(i)}, s_k^{(i)}] = \{s_1^{(i)}, \dots, s_{k-1}^{(i)}\}$.

Meanwhile, for the sentence $s_k^{(i)}$ in which character $e_j^{(i)}$ appears, check whether there are

motivations and emotions marked. If there are, the labels of this character in this sentence are

Maslow motivation tag $M[e_j^{(i)}, s_k^{(i)}] \subseteq M$, Reiss motivation tag $R[e_j^{(i)}, s_k^{(i)}] \subseteq R$ and Plutchik

emotion tag $P[e_j^{(i)}, s_k^{(i)}] \subseteq P$.

First, consider each tag in the Maslow category ($m_l \in M$). For every character in every story, if the sentence $s_k^{(i)}$ is the last sentence that has tag $m_l \in M[e_j^{(i)}, s_k^{(i)}]$, adding the combination of

the related character, the sentence, and all sentences before it, that is

$(e_j^{(i)}, s_k^{(i)}, C[e_j^{(i)}, s_k^{(i)}]) = (e_j^{(i)}, \{s_1^{(i)}, \dots, s_k^{(i)}\})$, to the text set corresponding to m_l . Finally, we get five character-story text collections corresponding to five tags respectively.

Then, consider each tag in the Reiss category ($r_t \in R$). For every character in every story, if the sentence $s_k^{(i)}$ is the last sentence that has tag $r_t \in R[e_j^{(i)}, s_k^{(i)}]$, adding the combination of the related character, the sentence, and all sentences before it, that is

$(e_j^{(i)}, s_k^{(i)}, C[e_j^{(i)}, s_k^{(i)}]) = (e_j^{(i)}, \{s_1^{(i)}, \dots, s_k^{(i)}\})$, to the text set corresponding to r_t . We get nineteen character-story text collections corresponding to nineteen tags respectively.

In the end, consider each tag in the Plutchik category ($p_n \in P$). For every character in every story, if the sentence $s_k^{(i)}$ is the last sentence that has tag $p_n \in P[e_j^{(i)}, s_k^{(i)}]$, adding the combination of the related character, the sentence, and all sentences before it, that is

$(e_j^{(i)}, s_k^{(i)}, C[e_j^{(i)}, s_k^{(i)}]) = (e_j^{(i)}, \{s_1^{(i)}, \dots, s_k^{(i)}\})$, to the text set corresponding to p_n . Finally, we get eight character-story text collections corresponding to eight tags respectively.

3.1.3 Encoding the corresponding texts of tags

Because in the text collection of the same label, different characters may come from the same story, their corresponding sentence sets overlap, and there is redundancy in coding. Therefore, the sets of sentences of different characters in the same story are unioned, so that there is only one set of sentences for each story belonging to each labeled text set. Concatenate the sentences into a text to obtain the final text set corresponding to each label of each category ($T^{m_l}, T^{r_t}, T^{p_n}$).

For the text set corresponding to each label m_l in the Maslow category (T^{m_l}), use a sentence encoder to encode each text $t_a^{m_l} \in T^{m_l}$ in it, then average the encodings of all $N_{T^{m_l}}$ texts to get the encoding corresponding to the label.

$$\text{Embed}[m_l] = \frac{\sum_{a=1}^{N_{T^{m_l}}} \text{Encoder}(t_a^{m_l})}{N_{T^{m_l}}} \quad (1)$$

For the Reiss category and the Plutchik category, use the same method separately to get the corresponding encoding for each motivation tag r_t and emotion tag p_n .

$$\text{Embed}[r_t] = \frac{\sum_{a=1}^{N_{T^{r_t}}} \text{Encoder}(t_a^{r_t})}{N_{T^{r_t}}} \quad (2)$$

$$\text{Embed}[p_n] = \frac{\sum_{a=1}^{N_{T^{p_n}}} \text{Encoder}(t_a^{p_n})}{N_{T^{p_n}}} \quad (3)$$

3.1.4 Encoding and annotating user texts

Users in the social media dataset are divided into two broad categories in terms of mental state: depressed and non-depressed. For each user u_i in each mental state, use a sentence encoder to obtain the user's text embedding $\text{Embed}[u_i]$. For each motivation and emotion category, compute the cosine similarity between the user's embedding and each tag encoding.

For each tag $m_l \in M$ in the Maslow category, calculate the probability that the user fits this tag.

$$\text{prob}[u_i, m_i] = \text{cos_sim}(\text{Embed}[u_i], \text{Embed}[m_i]) \quad (4)$$

For each tag $r_t \in R$ in the Reiss category, calculate the probability that the user fits this tag.

$$\text{prob}[u_i, r_t] = \text{cos_sim}(\text{Embed}[u_i], \text{Embed}[r_t]) \quad (5)$$

For each tag $p_n \in P$ in the Plutchik category, calculate the probability that the user fits this tag.

$$\text{prob}[u_i, p_n] = \text{cos_sim}(\text{Embed}[u_i], \text{Embed}[p_n]) \quad (6)$$

3.2 Annotation based on prompt

Directly prompt an interactive large language model to obtain the tags corresponding to the user text. The prompting method used is as follows:

对于输入文本“(input user text)”的内容, 请回答:

1. 文本表达了什么动机, 在['physiological', 'stability', 'love', 'esteem', 'spiritual growth']中选择*i* 个最符合的;

2. 文本表达了什么动机, 在['food', 'rest', 'health', 'savings', 'order', 'tranquility', 'romance', 'belonging', 'family', 'contact', 'competition', 'honor', 'approval', 'power', 'status', 'curiosity', 'serenity', 'idealism', 'independence']中选择*i* 个最符合的;

3. 文本表达了什么情绪, 在['joy', 'trust', 'surprise', 'anticipation', 'fear', 'disgust', 'anger', 'sadness']中选择*j* 个最符合的。

Because the literal meaning of tags is relatively intuitive, no explanation or examples of each tag are given, trying to observe how well LLM understands the psychological state implicit in the text in a natural state.

4 Experiments and Results

4.1 Datasets

The motivation and emotion data come from the storycommonsense dataset (Rashkiny et al., 2018). The original dataset is divided into a training set, a test set, and a development set, including 9885, 2370, and 2483 instances respectively. Each instance is a common sense story consisting of five sentences. All three sets have human-generated open-text descriptions of motivations and emotional responses for every sentence in which every character appears in each story, while only the test and dev set instances received theoretical category labels for both motivation and emotion, so only instances from these two sets are used in the experiments. In addition, for the Reiss category, the "tranquility" and "na" tags are additionally included in the data set and are also included during processing.

The social media data comes from the Identifying-depression dataset (Pirina and Çöltekin, 2018). Data comes from depression support and non-depression forums, as well as depression support and non-depression support subreddits on Reddit. We use four subsets as follows:

1) blogs_depression, contains 390 instances, the longest instance includes 5717 tokens, and the shortest instance includes 445 tokens.

2) blogs_non_depression, contains 933 instances, the longest instance includes 2445 tokens, and the shortest instance includes 54 tokens.

3) reddit_depression, contains 1293 instances, the longest instance includes 13874 tokens, and the shortest instance includes 12 tokens.

4) reddit_non_depression, contains 548 instances, the longest instance includes 17713 tokens,

and the shortest instance includes 12 tokens.

4.2 Data encoding

We use the uniform sentence encoder Jina Embeddings (Günther et al., 2023) through API provided by Jina AI. Input text into the jina-embeddings-v2-base-en model and get the 768-dimensional vector corresponding to the text.

For each tag in the Maslow, Reiss, and Plutchik category, 100 representative texts are randomly selected from their corresponding text sets (50 each for the test set and development set), use the Jina Embeddings to get the vector of each text, and then average 100 vectors as the vector corresponding to the tag.

For subsets `blogs_depression` and `blogs_non_depression`, randomly select 100 instances each from instances with no more than 1000 tokens each. For subsets `reddit_depression` and `reddit_non_depression`, randomly select 200 instances each from instances with no more than 1000 tokens each. Get the corresponding Jina embedding for each selected instance in all subsets.

4.3 Annotation based on similarity

For each selected user instance in all social media subsets, calculate the cosine similarity between the user embedding and each tag embedding in each category.

Since a user may have multiple motivations and multiple emotions at the same time, the three motivation labels with the highest similarity in the two motivation categories are selected for the user, and the four emotion labels with the highest similarity in the emotion category are selected for the user.

4.4 Annotation based on prompt

Select the top 30 of each of the 200 `reddit_depression` instances and 200 `reddit_non_depression` instances collected in Section 4.2 for annotation. Use the “文心一言” LLM from Baidu and the prompt in Section 3.2 to get motivation and emotion tags for each instance. The number of motivation tags $i=3$, and the number of emotion tags $j=4$.

4.5 Results

For the motivation and emotion tags of social media users obtained by the above two methods, for different social media sources, the total number of each motivation and emotion tag appearing in the depression category and non-depression category is counted separately. Because no user is assigned “safety” motivation, this tag is not displayed in the results. We compare the motivation and emotional differences between depressed and non-depressed users. The results are as follows.

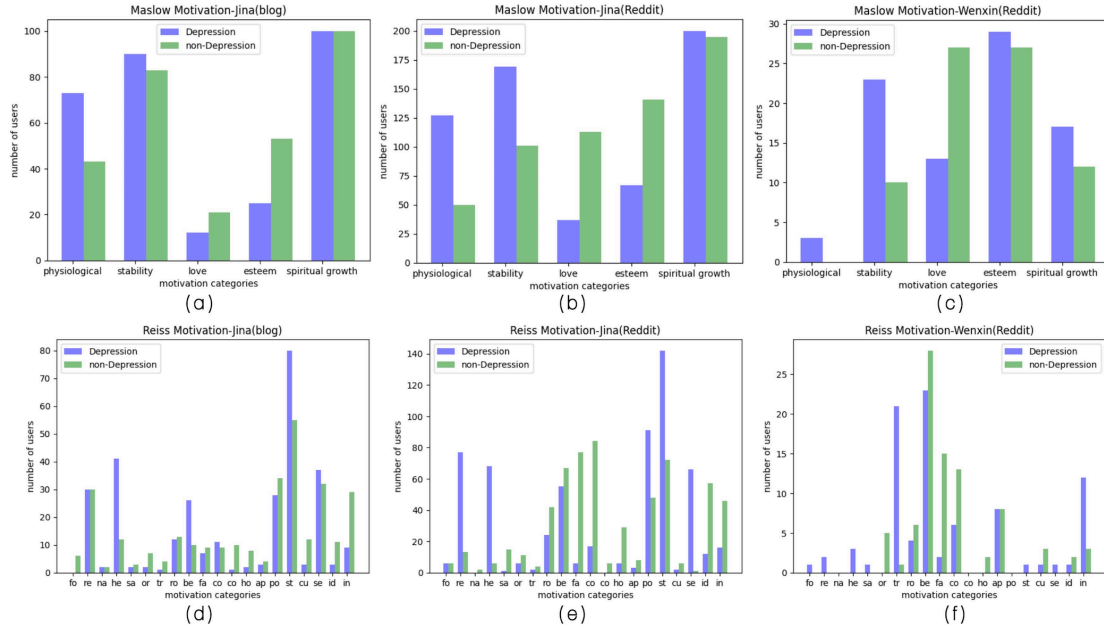


Figure 3: Motivation statistics after labeling social media data samples. Reiss tags only display the first two letters of the name for ease of display. (a) Maslow motivation statistics in depressed and non-depressed blog samples based on Jina embedding; (b) Maslow motivation statistics in depressed and non-depressed Reddit samples based on Jina embedding; (c) Maslow motivation statistics in depressed and non-depressed Reddit samples based on “文心一言”; (d) Reiss motivation statistics in depressed and non-depressed blog samples based on Jina embedding; (e) Reiss motivation statistics in depressed and non-depressed Reddit samples based on Jina embedding; (f) Reiss motivation statistics in depressed and non-depressed Reddit samples based on “文心一言”.

Observe the subgraphs obtained after calculating the similarity with Jina embedding in Figure 3, for different social media data sources, the motivation differences between depressed and non-depressed people are shown in Table 1. [] are the first three letters of the corresponding Maslow tags.

Table 1: Comparison of motivations between depressed and non-depressed social media data samples based on Jina embedding.

subgraphs	mental state	similarities	differences	
			blog	Reddit
(a),(b) (Maslow)	depressed	(most)spiritual growth (more)stability, physiological		
	non-depressed	(most)spiritual growth	(more)stability	(more)esteem, love
(d),(e) (Reiss)	depressed	(most)status[est] (more)health[sta] (many)serenity, rest[spi,phy]	(more)belonging[lov]	(more)power[est]
	non-depressed	(more)independence[spi]	(most)status[est] (many)serenity,	(most)contact, family[lov]

			rest[spi,phy]	(more)belonging[lov] (more)idealism[spi]
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When using the Jina encoder, it can be seen from the annotation results of Maslow motivation that people who use social media generally seek spiritual growth. Depressed people have more physiological motivations than non-depressed people. And this can be explained by the fact that depressed people may experience more physical symptoms. People who post on blogs are more likely to seek stability, while those who post on Reddit are more likely to seek respect and love. From the annotation results of Reiss motivation, we can see that basically, it is consistent with Maslow motivation. In addition to depressed people paying the most attention to status (i.e., respect in Maslow), depressed people on blogs will have more belonging motivations (i.e., love in Maslow).

Observe the subgraphs obtained by using two different annotation methods, the motivation differences between depressed and non-depressed people are shown in Table 2. [] are the first three letters of the corresponding Maslow tags.

Table 2: Comparison of motivations between depressed and non-depressed social media data samples based on two annotation methods.

subgraphs	mental state	similarities	differences	
			Jina embedding	文心一言(Wenxin)
(b),(c)	depressed	(more)stability	(most)spiritual growth (more)physiological	(most)esteem
	non-depressed		(most)spiritual growth (more)esteem	(most)esteem, love
(e),(f)	depressed		(most)status[est] (more)power, rest, health, serenity[est, phy, sta, spi]	(most)belonging, tranquility[lov, spi] (more)independence[spi] (many)approval[est]
	non-depressed	(more)contact, family[lov]	(more)idealism, independence[spi]	(more)belonging[lov] (many)approval[est]

From the labeling results of Maslow motivations for the two methods, we can see that non-depressed people on Reddit are relatively less stability-seeking. Wenxin assigns a high respect motivation to depressed people. Reiss motivation is almost consistent with Maslow motivation. Wenxin assigns the most belonging and tranquility to depressed people, rather than the status motivation obtained by Jina embedding. Jina embedding assigns more independent motivation to non-depressed people, while Wenxin assigns more independent motives to depressed people. Wenxin also assigns more approval motives to both groups of people on Reddit.

It can be seen that the large language model has identified high-status motivations from the texts of depressed people. Depression is a response to loss, failure, and helplessness, and may reflect the loss of social status of this group of people and their strong motivation to recover.

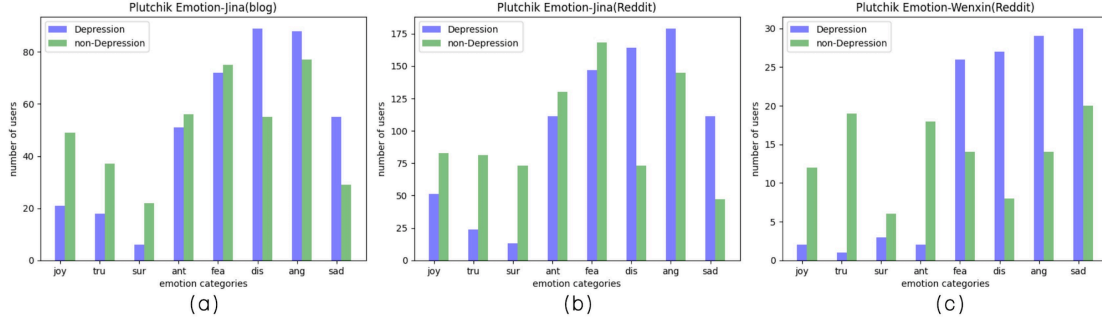


Figure 4: Emotion statistics after labeling social media data samples. Plutchik tags only display the first three letters of the name for ease of display. (a) Plutchik emotion statistics in depressed and non-depressed blog samples based on Jina embedding; (b) Plutchik emotion statistics in depressed and non-depressed Reddit samples based on Jina embedding; (c) Plutchik emotion statistics in depressed and non-depressed Reddit samples based on “文心一言”.

Observe the subgraphs obtained after calculating the similarity with Jina embedding in Figure 4, for different social media data sources, the emotion differences between depressed and non-depressed people are shown in Table 3.

Table 3: Comparison of emotions between depressed and non-depressed social media data samples based on Jina embedding.

subgraphs	mental state	similarities	differences	
			blog	Reddit
(a),(b)	depressed	(most)disgust, anger (many)anticipation, sad		
	non-depressed	(more)anticipation, joy, trust	(most)anger (many)fear	(most)fear (more)surprise

Observe the subgraphs obtained by using two different annotation methods, the emotion differences between depressed and non-depressed people are shown in Table 4.

Table 4: Comparison of emotions between depressed and non-depressed social media data samples based on two annotation methods.

subgraphs	Mental state	similarities	differences	
			Jina embedding	文心一言 (Wenxin)
(b),(c)	depressed	(many)disgust, fear	(most)anger (many)anticipation, joy	(most)sad
	non-depressed	(more)anticipation, trust, joy	(most)fear (more)surprise	(most)sad, trust

The depressed group reports significantly more disgust than the non-depressed group, possibly because they are more likely to report symptoms in disgusting situations, and depression is a cognitive disorder with very strong emotional symptoms (Grahek et al., 2019). The depressed

group also has a lot of expected emotions, probably because posting online itself is a kind of help-seeking behavior. Such depressed people have the motivation to improve themselves, and appropriate intervention can be considered. Furthermore, neither method assigns much anger to non-depressed users on Reddit, while non-depressed users on blogs get more anger tags, but both assign more trust to non-depressed users. Wenxin also assigns more sad emotions to non-depressed users.

These differences in motivation and emotion distribution may also arise in part from the themes of the data sources, i.e., differences in the data context.

5 Conclusions

According to the motivation tagging results obtained by using Jina embedding to calculate the similarity in Table 1, we can see that Jina embedding can express the corresponding relationship between the two categories of motivation and can show some differences in motivation between depressed and non-depressed people, as well as differences brought by certain text sources. The emotion annotation results in Table 3 also show the ability of Jina embedding to represent text semantics, which is basically in line with the general understanding of the emotional differences between the two types of people.

According to the results in Table 2 and Table 4, using a large language model for encoding and calculating similarity with the labeled data set, or directly using a large model without labeling the data set, can both obtain annotations that appear to be in line with general cognition, but there is also slight difference. In experiments, when using a large model to directly generate annotations, it will be found that sometimes the model directly gives additional annotations that are not included in the label set, such as the "safety" motivation that was not annotated in the common sense data set, and the "escape" motivation. This will be more flexible than a fixed set of labels. However, when more instances are asked in the same form, there may also be a phenomenon of answer convergence, which may not be due to the performance of the model itself, but to the use of permissions. This is also the main concern when applying the general large language model directly to the target task.

It can be seen that large language models can roughly identify the motivations and emotional states expressed in the text without domain adaptation training or without more detailed hints, which may be partly due to the unique nature of the mental health field itself. People's daily language itself has the purpose of communication and expression and is a partial manifestation of the cognitive process, it can reflect basic psychological characteristics to a certain extent. Therefore, large language models trained on massive amounts of diverse text can learn language patterns that express these features.

Measurement of mental health phenomena is largely based on subjective clinical impressions, and individual data are dominated by descriptive symptoms. Therefore, a diagnosis made using either of the existing disease classification systems (DSM-5 and ICD-11) cannot be used as a specific mental health status, but as a label that represents a heterogeneous group of diseases with similar phenotypic expressions, although these classification systems still serve as the gold standard for clinical diagnosis. The National Institute of Mental Health (NIMH) has also developed a new research framework, Research Domain Criteria (RDoC), to promote research into many dimensions of mental health and disease, shifting the attention of research from heterogeneous diagnostic

categories to specific psychological, emotional, behavioral and social constructs (Beyer and Boazak, 2021).

This work explores the differences in motivation and emotion distribution between self-reported depressed and non-depressed people on social media and only observes one of the roles that large language models can play when trying to model mental health-related constructs from individual language content. It can be used as a simple preliminary tagging method.

Because the judgment of an individual's mental state, such as motivation and emotion, is mostly a subjective impression, and an individual may have multiple motivations and emotions at the same time. Therefore, when trying to analyze an individual's motivations and emotions to promote improvement of mental health and avoid unhealthy behaviors, rather than focusing on performance on classification tasks, how to continuously obtain individual longitudinal data, and identify individual motivational and emotional processes to find cognitive-related defects, and then guide individuals to improve the related cognitive process to achieve better adaptability, which is of more practical significance.

6 Limitations and Future Directions

The limitations of this work mainly include the following aspects:

- 1) The dataset used is a published public dataset, and the two datasets are of different types. The corresponding data has not been specifically collected and labeled for the research question.
- 2) Only part of the data was used in the experiment, which is limited in data volume. There are other data resources in the original data set, which may be used in subsequent research.
- 3) Only one encoder was used as a preliminary observation of the ability of universal large language models in mental state modeling applications, lacking comparison with other modeling methods, which may be content for subsequent research.
- 4) The motivation and emotion distribution of different groups given by the language model has only been observed through preliminary experiments. The cognitive process of motivation and emotion in individual longitudinal data has not yet been modeled and analyzed. This may be the content of subsequent research.
- 5) The practical method of daily intervention based on cognitive process improvement combined with social media has not yet been explored, which may be included in subsequent research.

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